

## **Git and Github-Theoretical Test**

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Q1) Ans:- Git is a version control system, used to track all the changes made to our source code during software development. It helps multiple developers to collaborate, develop software, and maintain the code integrity throughout the development lifecycle.

Q2) Ans:- Git pull and Git fetch, both of them fetch the updates from the remote repo, but git pull automatically integrates them into our local branch, whereas git fetch doesn't, allowing review before merging.

Q3) Ans:- There are two ways to revert a commit:-

- Using git revert <commit\_hash> to create a new commit which undoes the changes of the commit whose hash is given by the user.
- Using git reset --hard <commit\_hash> to reset the repo to a previous commit, whose hash is given by the user, thus discarding the subsequent changes.

Q4) Ans:- Git staging area is the stage where the files, in which changes have been made, are kept before committing them to our repo. It allows us to review our changes before finalizing them. We can add unstaged files to staging area by git add <filename>

Q5)Ans:- A merge conflict is a case which occurs when Git is unable to reconcile the differences between two conflicting branches. It happens, when in two different branches ,the same lines of code are edited, or one branch edits the code and in the other branch, the file is deleted.

To resolve merge conflict, developers must manually choose which changes to keep, then mark them as resolved with git add, and then commit.

Q6) Ans:- Branching in Git allows multiple developers to work on separate features simultaneously, without impacting the main codebase. The branches can later be merged to the main branch, once the changes are completed and reviewed.

Q7)Ans:- Git rebase is a command to integrate changes from one branch onto another branch. It rewrites the commit history of one branch over another branch, as if they were originally made on top of it. It helps in maintaining a cleaner and linear project history.

Q8) Ans:-Git clone creates a copy of a remote repo on github, on our local system. Whereas, git fork creates a copy of a remote repo on github, on github itself.

Q9) Ans:- To delete a branch:-

- Use `git branch -d <branch_name>`, if the specified branch is fully merged into our current branch.
- Use `git branch -D <branch_name>` to delete forcefully, if the specified branch is not yet merged into our current branch.

Q10) Ans:- Git hook is an executable script which is automatically executed by Git, before or after specific Git events, such as commit, merge, and push. It allows us to customize and automate actions in our workflow.

It can be used in the following ways:-

- To execute code formatting checks, to ensure code quality before commit ( Pre- commit)
- To validate incoming pushes to a remote repo(Pre- receive), like rejecting commits which don't adhere to specific guidelines.
- To trigger actions after a push to a remote repo(Post – receive), like notifications.
- To validate outgoing pushes to a remote repo( Pre- push), like running tests to prevent pushing incomplete and broken changes.
- To modify commit messages automatically before they are finalized(Prepare-commit-msg), such as adding issue numbers or tags based on the branch name.